

$$3d^6$$

August 18, 2026

$$1 \quad \left| 3d_{xz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha 3d_{yz}^0 \right\rangle$$

Here are the CSFs:

$$|1, 2, 2\rangle = \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|2, 2, 2\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|3, 2, 2\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|4, 2, 2\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|5, 2, 2\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle$$

$$\begin{aligned} |6, 2, 1\rangle = & \frac{1}{2} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\ & + \frac{1}{2} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{2} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \end{aligned}$$

$$\begin{aligned} |7, 2, 1\rangle = & \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\ & + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \end{aligned}$$

$$\begin{aligned} |8, 2, 1\rangle = & \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\ & + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \end{aligned}$$

$$\begin{aligned} |9, 2, 1\rangle = & \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\ & + \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \end{aligned}$$

[illegible]

$$\begin{aligned}
|17, 2, -1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|18, 2, -1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|19, 2, -1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|20, 2, -1\rangle &= \frac{1}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|21, 2, -2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|22, 2, -2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|23, 2, -2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|24, 2, -2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|25, 2, -2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|26, 1, 1\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|27, 1, 1\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|28, 1, 1\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|29, 1, 1\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|30, 1, 1\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|31, 1, 0\rangle &= +\frac{1}{2} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{2} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{2} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{2} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

[illegible]

5

$$\begin{aligned}
|51, 1, -1\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|52, 1, -1\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|53, 1, -1\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|54, 1, -1\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|55, 1, -1\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|56, 1, 1\rangle &= \frac{\sqrt{3}}{2} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{12}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|57, 1, 1\rangle &= \frac{\sqrt{3}}{2} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|58, 1, 1\rangle &= \frac{\sqrt{3}}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|59, 1, 1\rangle &= \frac{\sqrt{3}}{2} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle
\end{aligned}$$

[illegible]

$$\begin{aligned}
|67, 1, -1\rangle &= \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{2} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|68, 1, -1\rangle &= \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|69, 1, -1\rangle &= \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{\sqrt{3}}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|70, 1, -1\rangle &= \frac{1}{\sqrt{12}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{12}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{\sqrt{3}}{2} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|71, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|72, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|73, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|74, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|75, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|76, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|77, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|78, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|79, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|80, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|81, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|82, 1, 1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|83, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|84, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|85, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|86, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|87, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|88, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|89, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|90, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|91, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|92, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|93, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|94, 1, 1\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|95, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|96, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|97, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|98, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|99, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|100, 1, 1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|101, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|102, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

[illegible]

$$\begin{aligned}
|121, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|122, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|123, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|124, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|125, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|126, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|127, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|128, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|129, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|130, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|131, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|132, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|133, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|134, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|135, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|136, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|137, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|138, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|139, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|140, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|141, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|142, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|143, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\
|144, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|145, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|146, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|147, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|148, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|149, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|150, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|151, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|152, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|153, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|154, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|155, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|156, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|157, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|158, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|159, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|160, 1, -1\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

[illegible]

[illegible]

$$\begin{aligned}
|201, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\
|202, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|203, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|204, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|205, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|206, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|207, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|208, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|209, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|210, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

The Ground State CSFs are 80 110 140

$$\begin{aligned}
\langle 80, 1, 1 | L_x | 74, 1, 1 \rangle &= -I \\
\langle 80, 1, 1 | L_x | 75, 1, 1 \rangle &= \sqrt{3}I
\end{aligned}$$

$$\Delta g_{xx}$$

$$\begin{aligned}
&+ \zeta \Delta_{80-74}^{-1} \\
&+ 3\zeta \Delta_{80-75}^{-1}
\end{aligned}$$

$$\begin{aligned}
\langle 80, 1, 1 | L_y | 26, 1, 1 \rangle &= \sqrt{2}I \\
\langle 80, 1, 1 | L_y | 92, 1, 1 \rangle &= -\sqrt{3}I \\
\langle 80, 1, 1 | L_y | 93, 1, 1 \rangle &= -I
\end{aligned}$$

$$\Delta g_{yy}$$

$$\begin{aligned}
&+ 2\zeta \Delta_{80-26}^{-1} \\
&+ 3\zeta \Delta_{80-92}^{-1} \\
&+ \zeta \Delta_{80-93}^{-1}
\end{aligned}$$

$$\begin{aligned}\langle 80, 1, 1 | L_z | 28, 1, 1 \rangle &= -\sqrt{2}I \\ \langle 80, 1, 1 | L_z | 82, 1, 1 \rangle &= 2I\end{aligned}$$

$$\Delta g_{zz}$$

$$\begin{aligned}&+2\zeta\Delta_{80-28}^{-1} \\ &+4\zeta\Delta_{80-82}^{-1}\end{aligned}$$

For a multiplicity of 3 we have the following:

$$D = \langle 1, 1 | 1, 1 \rangle - \langle 1, 0 | 1, 0 \rangle$$

$$E = \langle 1, -1 | 1, 1 \rangle$$

$$\begin{aligned}\langle 1, 1 | \hat{H}_{eff} | 1, 1 \rangle &= -\zeta^2 (\\ &+ \frac{1}{4}\Delta_1^{-1} + \frac{1}{4}\Delta_8^{-1} + \frac{1}{24}\Delta_{11}^{-1} + \frac{1}{6}\Delta_{43}^{-1} + \frac{1}{12}\Delta_{46}^{-1} + \frac{1}{12}\Delta_{58}^{-1} \\ &+ \frac{1}{24}\Delta_{61}^{-1} + \Delta_{82}^{-1} + \frac{1}{8}\Delta_{104}^{-1} + \frac{3}{8}\Delta_{105}^{-1} + \frac{3}{8}\Delta_{122}^{-1} + \frac{1}{8}\Delta_{123}^{-1} \\ &+ \frac{1}{12}\Delta_{166}^{-1} + \frac{1}{8}\Delta_{174}^{-1} + \frac{3}{8}\Delta_{175}^{-1} + \frac{3}{8}\Delta_{192}^{-1} + \frac{1}{8}\Delta_{193}^{-1})\end{aligned}$$

$$\begin{aligned}\langle 1, 0 | \hat{H}_{eff} | 1, 0 \rangle &= -\zeta^2 (\\ &+ \frac{1}{8}\Delta_6^{-1} + \frac{1}{3}\Delta_{13}^{-1} + \frac{1}{8}\Delta_{16}^{-1} + \frac{1}{12}\Delta_{41}^{-1} + \frac{1}{12}\Delta_{51}^{-1} + \frac{1}{24}\Delta_{56}^{-1} \\ &+ \frac{1}{24}\Delta_{66}^{-1} + \frac{1}{8}\Delta_{74}^{-1} + \frac{3}{8}\Delta_{75}^{-1} + \frac{3}{8}\Delta_{92}^{-1} + \frac{1}{8}\Delta_{93}^{-1} + \frac{1}{8}\Delta_{134}^{-1} \\ &+ \frac{3}{8}\Delta_{135}^{-1} + \frac{3}{8}\Delta_{152}^{-1} + \frac{1}{8}\Delta_{153}^{-1} + \frac{1}{6}\Delta_{168}^{-1} + \Delta_{182}^{-1})\end{aligned}$$

$$\begin{aligned}\langle 1, -1 | \hat{H}_{eff} | 1, 1 \rangle &= -\zeta^2 (\\ &+ \frac{1}{24}\Delta_{11}^{-1} - \frac{1}{12}\Delta_{46}^{-1} - \frac{1}{24}\Delta_{61}^{-1} + \frac{1}{8}\Delta_{104}^{-1} + \frac{3}{8}\Delta_{105}^{-1} \\ &- \frac{3}{8}\Delta_{122}^{-1} - \frac{1}{8}\Delta_{123}^{-1} + \frac{1}{12}\Delta_{166}^{-1} - \frac{1}{8}\Delta_{174}^{-1} - \frac{3}{8}\Delta_{175}^{-1} \\ &+ \frac{3}{8}\Delta_{192}^{-1} + \frac{1}{8}\Delta_{193}^{-1})\end{aligned}$$

$$\begin{aligned}
D = & \zeta^2 (\\
& -\frac{1}{4}\Delta_1^{-1} + \frac{1}{8}\Delta_6^{-1} - \frac{1}{4}\Delta_8^{-1} - \frac{1}{24}\Delta_{11}^{-1} + \frac{1}{3}\Delta_{13}^{-1} + \frac{1}{8}\Delta_{16}^{-1} \\
& + \frac{1}{12}\Delta_{41}^{-1} - \frac{1}{6}\Delta_{43}^{-1} - \frac{1}{12}\Delta_{46}^{-1} + \frac{1}{12}\Delta_{51}^{-1} + \frac{1}{24}\Delta_{56}^{-1} - \frac{1}{12}\Delta_{58}^{-1} \\
& - \frac{1}{24}\Delta_{61}^{-1} + \frac{1}{24}\Delta_{66}^{-1} + \frac{1}{8}\Delta_{74}^{-1} + \frac{3}{8}\Delta_{75}^{-1} - \Delta_{82}^{-1} + \frac{3}{8}\Delta_{92}^{-1} \\
& + \frac{1}{8}\Delta_{93}^{-1} - \frac{1}{8}\Delta_{104}^{-1} - \frac{3}{8}\Delta_{105}^{-1} - \frac{3}{8}\Delta_{122}^{-1} - \frac{1}{8}\Delta_{123}^{-1} + \frac{1}{8}\Delta_{134}^{-1} \\
& + \frac{3}{8}\Delta_{135}^{-1} + \frac{3}{8}\Delta_{152}^{-1} + \frac{1}{8}\Delta_{153}^{-1} - \frac{1}{12}\Delta_{166}^{-1} + \frac{1}{6}\Delta_{168}^{-1} - \frac{1}{8}\Delta_{174}^{-1} \\
& - \frac{3}{8}\Delta_{175}^{-1} + \Delta_{182}^{-1} - \frac{3}{8}\Delta_{192}^{-1} - \frac{1}{8}\Delta_{193}^{-1})
\end{aligned}$$

$$\begin{aligned}
E = & \zeta^2 (\\
& -\frac{1}{24}\Delta_{11}^{-1} + \frac{1}{12}\Delta_{46}^{-1} + \frac{1}{24}\Delta_{61}^{-1} - \frac{1}{8}\Delta_{104}^{-1} - \frac{3}{8}\Delta_{105}^{-1} \\
& + \frac{3}{8}\Delta_{122}^{-1} + \frac{1}{8}\Delta_{123}^{-1} - \frac{1}{12}\Delta_{166}^{-1} + \frac{1}{8}\Delta_{174}^{-1} + \frac{3}{8}\Delta_{175}^{-1} \\
& - \frac{3}{8}\Delta_{192}^{-1} - \frac{1}{8}\Delta_{193}^{-1})
\end{aligned}$$